

● HARD

● CRAFT AND STRUCTURE

Craft and Structure

10 questions · ~15 min

03

Text 1

Soy sauce, made from fermented soybeans, is noted for its umami flavor. Umami—one of the five basic tastes along with sweet, bitter, salty, and sour—was formally classified when its taste receptors were discovered in the 2000s. In 2007, to define the pure umami flavor scientists Rie Ishii and Michael O’Mahony used broths made from shiitake mushrooms and kombu seaweed, and two panels of Japanese and US judges closely agreed on a description of the taste.

Text 2

A 2022 experiment by Manon Jünger et al. led to a greater understanding of soy sauce’s flavor profile. The team initially presented a mixture of compounds with low molecular weights to taste testers who found it was not as salty or bitter as real soy sauce. Further analysis of soy sauce identified proteins, including dipeptides, that enhanced umami flavor and also contributed to saltiness. The team then made a mix of 50 chemical compounds that re-created soy sauce’s flavor.

Based on the texts, if Ishii and O’Mahony (Text 1) and Jünger et al. (Text 2) were aware of the findings of both experiments, they would most likely agree with which statement?

- A. On average, the diets of people in the United States tend to have fewer foods that contain certain dipeptides than the diets of people in Japan have.
- B. Chemical compounds that activate both the umami and salty taste receptors tend to have a higher molecular weight than those that only activate umami taste receptors.
- C. Fermentation introduces proteins responsible for the increase of umami flavor in soy sauce, and those proteins also increase the perception of saltiness.

- D. The broths in the 2007 experiment most likely did not have a substantial amount of the dipeptides that played a key part in the 2022 experiment.

The 1967 release of Harold Cruse’s book *The Crisis of the Negro Intellectual* isolated him from almost all other scholars and activists of the American Civil Rights Movement—though many of those thinkers disagreed with each other, he nonetheless found ways to disagree with them all. He thought that activists who believed that Black people such as himself should culturally assimilate were naïve. But he also sharply criticized Black nationalists such as Marcus Garvey who wanted to establish independent, self-contained Black economies and societies, even though Cruse himself identified as a Black nationalist.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It describes a direction that Cruse felt the Civil Rights Movement ought to take.
- B. It indicates that Cruse’s reputation as a persistent antagonist of other scholars is undeserved.
- C. It describes a controversy that Cruse’s work caused within the Black nationalist movement.
- D. It helps explain Cruse’s position with respect to the community of civil rights thinkers.

Containing over 160 billion base pairs of DNA resulting from polyploidy (whole genome duplication) and accumulation of noncoding DNA without significant deletion, the genome of *Tmesipteris oblongeolata* (New Caledonian fork fern) is the largest known among eukaryotes—and, given the high biological costs of extremely large genomes, is unlikely to be _____ blank.

Which choice completes the text with the most logical and precise word or phrase?

- A. protracted
- B. superseded
- C. curtailed
- D. obviated

Text 1

The fossil record suggests that mammoths went extinct around 11 thousand years (kyr) ago. In a 2021 study of environmental DNA (eDNA)—genetic material shed into the environment by organisms—in the Arctic, Yucheng Wang and colleagues found mammoth eDNA in sedimentary layers formed millennia later, around 4 kyr ago. To account for this discrepancy, Joshua H. Miller and Carl Simpson proposed that arctic temperatures could preserve a mammoth carcass on the surface, allowing it to leach DNA into the environment, for several thousand years.

Text 2

Wang and colleagues concede that eDNA contains DNA from both living organisms and carcasses, but for DNA to leach from remains over several millennia requires that the remains be perpetually on the surface. Scavengers and weathering in the Arctic, however, are likely to break down surface remains well before a thousand years have passed.

Which choice best describes how Text 1 and Text 2 relate to each other?

- A. Text 1 discusses two approaches to studying mammoth extinction without advocating for either, whereas Text 2 advocates for one approach over the other.
- B. Text 1 presents findings by Wang and colleagues and gives another research team's attempt to explain those findings, whereas Text 2 provides additional detail that calls that explanation into question.
- C. Text 1 describes Wang and colleagues' study and a critique of their methodology, whereas Text 2 offers additional details showing that methodology to be sound.
- D. Text 1 argues that new research has undermined the standard view of when mammoths went extinct, whereas Text 2 suggests a way to reconcile the standard view with that new research.

Several studies have found negligible electoral consequences for governments that impose fiscal austerity measures, yet some European governments recently suffered electorally due to their austerity programs. Evelyne Huebscher and colleagues attribute this incongruity to governments' tendency—not followed in the recent European cases—to implement austerity programs strategically to avoid electoral costs (e.g., setting spending cuts to take effect only after the next election), which has obscured the inherent political risks of austerity measures in the election data scholars have examined.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It explains a discrepancy between what has been observed in study settings and what has been observed in real-world settings that the text goes on to assert is attributable to the studies not using real-world data.
- B. It identifies a conflict between research findings and recent events that the text goes on to suggest is a consequence of a complicating factor in the data used to generate those findings.
- C. It presents a long-standing divergence in research findings that the text goes on to say is due to different groups of researchers using data that derive from different electoral circumstances.
- D. It describes a recent exception to a general pattern in research findings that the text goes on to explain is a result of researchers underestimating the significance of inconsistencies in the data they've analyzed.

Barring major archaeological discoveries, we are unlikely to ever have _____ blank account of ancient Egypt under the female pharaoh Hatshepsut, as much of the evidence of her reign was deliberately destroyed by her successors.

Which choice completes the text with the most logical and precise word or phrase?

- A. an imaginative
- B. a superficial
- C. an exhaustive
- D. a questionable

Text 1

In 1916, H. Dugdale Sykes disputed claims that *The Two Noble Kinsmen* was coauthored by William Shakespeare and John Fletcher. Sykes felt Fletcher's contributions to the play were obvious—Fletcher had a distinct style in his other plays, so much so that lines with that style were considered sufficient evidence of Fletcher's authorship. But for the lines not deemed to be by Fletcher, Sykes felt that their depiction of women indicated that their author was not Shakespeare but Philip Massinger.

Text 2

Scholars have accepted *The Two Noble Kinsmen* as coauthored by Shakespeare since the 1970s: it appears in all major one-volume editions of Shakespeare's complete works. Though scholars disagree about who wrote what exactly, it is generally held that on the basis of style, Shakespeare wrote all of the first act and most of the last, while John Fletcher authored most of the three middle acts.

Based on the texts, both Sykes in Text 1 and the scholars in Text 2 would most likely agree with which statement?

- A. John Fletcher's writing has a unique, readily identifiable style.
- B. The women characters in John Fletcher's plays are similar to the women characters in Philip Massinger's plays.
- C. *The Two Noble Kinsmen* belongs in one-volume compilations of Shakespeare's complete plays.
- D. Philip Massinger's style in the first and last acts of *The Two Noble Kinsmen* is an homage to Shakespeare's style.

The field of study called affective neuroscience seeks instinctive, physiological causes for feelings such as pleasure or displeasure. Because these sensations are linked to a chemical component (for example, the release of the neurotransmitter dopamine in the brain when one receives or expects a reward), they can be said to have a partly physiological basis. These processes have been described in mammals, but Jingnan Huang and his colleagues have recently observed that some behaviors of honeybees (such as foraging) are also motivated by a dopamine-based signaling process.

What choice best describes the main purpose of the text?

- A. It describes an experimental method of measuring the strength of physiological responses in humans.
- B. It illustrates processes by which certain insects can express how they are feeling.
- C. It summarizes a finding suggesting that some mechanisms in the brains of certain insects resemble mechanisms in mammalian brains.
- D. It presents research showing that certain insects and mammals behave similarly when there is a possibility of a reward for their actions.

Close analysis of the painting *Girl with a Flute*, long attributed to the seventeenth-century Dutch painter Johannes Vermeer, has revealed subtle deviations from the artist's signature techniques. These variations suggest that the work may be that of a student under Vermeer's tutelage—potentially _____ blank our understanding of Vermeer as a solitary artist.

Which choice completes the text with the most logical and precise word or phrase?

- A. negating
- B. prefiguring
- C. entrenching
- D. substantiating

Text 1

Like the work of Ralph Ellison before her, Toni Morrison's novels feature scenes in which characters deliver sermons of such length and verbal dexterity that for a time, the text exchanges the formal parameters of fiction for those of oral literature. Given the many other echoes of Ellison in Morrison's novels, both in structure and prose style, these scenes suggest Ellison's direct influence on Morrison.

Text 2

In their destabilizing effect on literary form, the sermons in Morrison's works recall those in Ellison's. Yet literature by Black Americans abounds in moments where interpolated speech erodes the division between oral and written forms that literature in English has traditionally observed. Morrison's use of the sermon is attributable not only to the influence of Ellison but also to a community-wide strategy of resistance to externally imposed literary conventions.

Based on the texts, how would the author of Text 2 most likely characterize the underlined claim in Text 1?

- A. As failing to consider Ellison's and Morrison's equivalent uses of the sermon within the wider cultural context in which they wrote
- B. As misunderstanding the function of sermons in novels by Black American writers other than Ellison and Morrison
- C. As disregarding points of structural and stylistic divergence between the works of Ellison and those of Morrison
- D. As being indebted to the tradition of resisting literary conventions that privilege written forms, such as novels, over sermons and other oral forms